# Chapter 6

# Seeds – Structure and Germination

*GO.*

## 📝 Part 1: In-Chapter Point-Wise Notes

### 🌱 6.1 What is a Seed?

- **Fruit:** The enlarged ripened ovary, where the ovarian wall forms the fruit wall that encloses and protects the seeds and helps in seed dispersal. ➡️ *(Example: Mango, pea pod).*
- **Seed:** The ripened ovule (a fertilized ovule develops into a seed). It contains an embryo which develops into a new plant. The seed coat protects the embryo from mechanical damage. ➡️ *(Example: Bean seed, peas).*
- **Grain:** Actually a fruit in which the fruit wall and the seed coat are completely fused together to form a protective layer. ➡️ *(Example: Maize, wheat, rice).*
- **More About the Seed:**
    - It is a mature ovule after fertilisation, containing a tiny living plant called the embryo (developed from the fused sperm nucleus and egg nucleus).
    - The embryo consists of a radicle (future root), a plumule (future shoot), and cotyledons (seed leaves).
    - The embryo remains in an inactive (dormant) state and germinates only upon exposure to favourable conditions.
    - Some seeds can withstand unfavourable conditions (drought, temperature) and remain dormant for 100 years or more.
    - The seed may also contain endosperm which, along with the cotyledons, stores food material for the nourishment of the embryo during germination.

### 🗂️ 6.2 Classification and Structure of Seeds

- **Size Variance:** Seeds can be barely visible to the naked eye (poppy, strawberry), quite large (mango, plum), or massive (coconut, double coconut being the largest).
- **Broad Classification based on Cotyledons:**
    - **Monocotyledonous:** Seeds containing one cotyledon. These plants typically have fibrous roots and parallel leaf venation. ➡️ *(Example: Maize, grasses).*
    - **Dicotyledonous:** Seeds containing two cotyledons. These plants typically have a tap root and reticulate leaf venation. ➡️ *(Example: Pea, gram, bean).*
- **Classification based on Endosperm:**

| **Parameter** 📏 | **Albuminous (Endospermic) Seeds** 🌾 | **Exalbuminous (Non-endospermic) Seeds** 🥜 |
|---|---|---|
| **Endosperm Status** | Endosperm persists to store food. | Endosperm is consumed; cotyledon stores the food and becomes thick and fleshy. |
| **Cotyledon Structure** | Cotyledons are thin and membranous; they **do not store food**. | Cotyledons are thick and fleshy due to stored food. |
| **Monocot Examples** | Cereals, millets, palm. | Vallisneria, orchids, amorphophallus. |
| **Dicot Examples** | Poppy, custard apple. | Gram, pea, mango, mustard. |

🫘 **1. The Bean Seed (Dicotyledonous)**

- Most beans are kidney-shaped with a convex and a concave side.

- **Seed Coat:** Protects the delicate inner parts from injury, bacteria, fungi, and insects. It consists of two layers:
    - **Testa:** The strong, outermost hard brownish covering.
    - **Tegmen:** A thin inner layer lying next to the testa, also protective in nature.
- **Hilum:** A distinct whitish oval scar on the concave side representing the spot where the ovule was attached to the ovary wall through the placenta.
- **Micropyle:** A tiny pore situated close to the hilum that marks the opening through which the pollen tube had entered the ovule.
    - ⚙️ **Function 1:** When soaked, seeds absorb water mainly through the micropyle, making it available to the embryo for germination.
    - ⚙️ **Function 2:** Provides for the diffusion of respiratory gases for the growing embryo.
- **The Embryo Axis:** Attached to the cotyledons.
    - **Radicle:** Forms the future root.
    - **Plumule:** Forms the future shoot, consisting of a short stem, a pair of tiny leaves, and a growing point.
    - **Epicotyl:** The region of the axis between the point of attachment of cotyledons and the plumule.
    - **Hypocotyl:** The region of the axis below the cotyledons.

### 🌽 2. Maize Grain (Monocotyledonous)

- The maize grain is a single-seeded fruit where the fruit wall and the seed coat are completely fused to form a single grain.
- A small light-coloured oval area on one side marks the location of the embryo, while the remaining major part contains a large starch-rich endosperm.
- **Epithelial Layer:** A thin layer separating the endosperm from the embryonic part.

- **Aleurone Layer:** The outermost layer of the endosperm which is highly rich in protein.
- **Scutellum:** The single, large cotyledon of the maize embryo.
- **Coleorhiza:** A protective sheath enclosing the radicle towards the pointed end.
- **Coleoptile:** A protective sheath enclosing the plumule towards the upper broader side.

| Parameter 📏 | Bean Seed 🫘 | Maize Grain 🌽 |
|---|---|---|
| **Cotyledons** | Two cotyledons. | One cotyledon (scutellum). |
| **Endosperm** | No endosperm. | Large endosperm present. |
| **Embryo Size** | Large embryo. | Small embryo. |
| **Plumule Leaves** | Plumule leaves folded; Plumule is large. | Plumule leaves rolled; Plumule is very small. |
| **Hilum & Micropyle** | Hilum and micropyle are distinctly visible. | Hilum and micropyle are not visible. |
| **Outer Covering** | Seed separately contained in the fruit called a pod. | Seed wall and fruit wall fused to form a single grain with no separate seed. |

## 🌱 6.3 Germination and Conditions Necessary

- **Dormancy:** A period of rest where the embryo is inactive. Though seemingly lifeless, dry seeds slowly consume minute quantities of oxygen, release carbon dioxide, and generate a little heat.
- **Germination:** The process of formation of a seedling developed from the embryo when placed under proper conditions. (Note: Fresh seeds must

first pass through a period of dormancy for physiological maturation before they can germinate).

- **Three Conditions Necessary for Germination:**
    1. **Water:** Absorbed from the soil mainly through the micropyle.
        - It causes the seed to swell and the seed coat to rupture, allowing the elongating radicle to come out and form the root system.
        - It provides the medium for chemical reactions, allowing enzymes to dissolve stored food in the cotyledons/endosperm into a diffusible form for the growing embryo.
    2. **Suitable (Optimum) Temperature:** Very low temperatures inhibit growth, while very high temperatures destroy delicate tissues. A moderately warm temperature of 25°C to 35°C is usually optimum.
    3. **Oxygen:** Required for respiration (oxidation of food) to provide the energy needed for rapid cell division and cell growth during germination.
- **Deep-Sown Seed Failure (Action Point):** Seeds sown very deep in soil fail to germinate because they lack a proper supply of oxygen for respiration, and their embryonic parts (hypocotyl/epicotyl) do not have sufficient pushing force to break through thick upper layers of soil.

## 📝 Part 1: In-Chapter Point-Wise Notes

### 🧪 6.4 Some Experiments on Germination

- **Experiment 1 (Proving water is necessary):** Take two beakers; place seeds on wet cotton wool in Beaker A and on dry cotton wool in Beaker B. Kept in an ordinary room, the seeds in Beaker A will germinate but not in Beaker B, proving water is necessary.
- **Experiment 2 (Proving suitable temperature is necessary):** Place seeds on wet cotton wool in two beakers; keep Beaker A in an ordinary room and Beaker B in a refrigerator. Seeds in Beaker A germinate, while seeds in

Beaker B show no signs of germination (or are heavily delayed), showing the importance of a suitable temperature.

- **Experiment 3 (Proving oxygen is necessary):**
    - Set up two flasks with soaked gram seeds on wet cotton wool.
    - In Flask A, hang a test tube containing plain water (which **maintains a humid environment**).
    - In Flask B, hang a test tube containing alkaline pyrogallic acid, which absorbs oxygen.
    - Seeds in Flask A germinate, but seeds in Flask B do not (except for very slight anaerobic respiration), proving oxygen is required.
- **Experiment 4 (The Three-Bean Seeds Experiment):** Three mature air-dried seeds are tied to a glass slide at three different positions in a beaker containing water.
    - **Top seed:** Kept well above water; it gets only oxygen but no water ➡️ Does not germinate at all.
    - **Middle seed:** Kept just at the water level; it gets both oxygen and water ➡️ Germinates successfully.
    - **Bottom seed:** Completely immersed in water; it gets water but very little oxygen (only from dissolved air) ➡️ Does not germinate or stops germinating after a small radicle emerges.
    - *Note:* This conclusively proves water is essential, but the exact requirement of oxygen is not fully demonstrated because the bottom seed still receives dissolved oxygen.

### 🌱 6.5 Types of Germination

- During germination, both the epicotyl and hypocotyl of a seed never elongate together; it is always either one or the other that elongates.

| Parameter 📏 | Hypogeal Germination ⬇️ | Epigeal Germination ⬆️ |
|---|---|---|
| **Name Meaning** | *hypo*: below + *geo*: earth. | *epi*: above + *geo*: earth. |
| **Elongating Part** | Epicotyl elongates faster. | Hypocotyl elongates faster. |
| **Cotyledon Position** | Cotyledons remain underground (or on the ground). | Cotyledons are pushed above the ground. |
| **Occurrence Table Fact** | Usually occurs in monocotyledonous seeds. | Usually occurs in dicotyledonous seeds. |
| **Inline Examples** | ➡️ *(Example: pea and gram).* | ➡️ *(Example: castor, bean).* |

## 🌿 6.6 Germination in Some Common Seeds

- **Pea Seed (Hypogeal Germination):**
    - The seed absorbs water and swells considerably, causing the testa to soften and burst.
    - The radicle emerges first, growing downwards to form the root system.
    - The plumule then grows upwards to form the shoot; in its early stages, it remains arched to protect the young shoot from injury as it pushes through the soil.
    - The cotyledons remain underground, supplying food until the seedling can exist independently, after which they wither and shrivel.
- **Bean Seed (Epigeal Germination):**

    - The seed absorbs water and swells, allowing the radicle to grow downwards to form the root system.
    - The arched hypocotyl grows, forming an arch/loop above the soil.
    - It then straightens, physically bringing and pushing the cotyledons above the soil.
    - The cotyledons become the first green leaves but soon fall off once the true foliage leaves grow.
- **Maize Grain (Hypogeal Germination):**
    - The grain imbibes water and swells.
    - The radicle pierces through the protective root sheath (coleorhiza) and the fruit wall to grow downwards and form the root system, but it dies off soon.
    - New roots, specifically called **adventitious roots**, develop from the base of the stem to take over.
    - The plumule pierces through its protective sheath (coleoptile) and grows straight upwards.
    - The single cotyledon (scutellum) absorbs food from the endosperm till it is exhausted; importantly, the hypocotyl does not elongate.
- **Viviparous Germination (Vivipary):**
    - A special mode of seed germination where the seed germinates *inside* the fruit while it is still attached to the parent plant.
    - The term *viviparous* literally translates to **producing live young ones**.
    - After germination, the plant drops the seedling directly into the soil/mud below, where it develops a root and fixes itself.
    - ➡️ *(Example: Mangrove plants like Rhizophora and Sonneratia).*

🪴 **The Seedling**

- Germination in any plant ends with the formation of a seedling.

- A seedling is defined as a stage in the growth of a plant from a seed before it has become wholly independent of the food stored in it.
- Eventually, the roots of the seedling absorb water and minerals from the soil, and the leaves start manufacturing food for the young plant, which keeps growing to become a mature plant.

## 🏆 Part 2: End-of-Chapter Master Lists (Seeds & Germination) 🌱

### 📚 1. Combined Terms List (Glossary) 🧠

| Biological Term 🔑 | Exam-Ready Definition 📝 |
|---|---|
| **Fruit** | The enlarged **ripened ovary**, whose wall encloses and protects the seeds. |
| **Seed** | The **ripened ovule** containing an embryo which develops into a new plant. |
| **Grain** | Actually a fruit in which the fruit wall and the seed coat are **completely fused** together to form a protective layer. |
| **Embryo** | A tiny living plant inside the seed developed from the **fused sperm and egg nucleus**. |
| **Radicle** | The part of the embryo that forms the **future root**. |
| **Plumule** | The part of the embryo that forms the **future shoot**. |
| **Dormancy** | A period of rest where the embryo remains in an **inactive state** until exposed to favourable conditions. |

| Biological Term 🔑 | Exam-Ready Definition 📝 |
| --- | --- |
| **Albuminous (Endospermic)** | Seeds in which the **endosperm persists** and cotyledons are thin and membranous. |
| **Exalbuminous (Non-endospermic)** | Seeds in which the endosperm is consumed and the **cotyledon stores food**, becoming thick and fleshy. |
| **Testa** | The outermost hard, brownish, **protective covering** of the seed coat. |
| **Tegmen** | The thin inner layer lying **next to the testa**, also protective in nature. |
| **Hilum** | A distinct whitish oval scar representing the spot where the ovule was **attached to the ovary wall**. |
| **Micropyle** | A tiny pore close to the hilum marking the opening through which the **pollen tube entered** the ovule. |
| **Epicotyl** | The region of the embryo axis **between** the point of attachment of cotyledons and the plumule. |
| **Hypocotyl** | The region of the embryo axis **below** the cotyledons. |
| **Aleurone Layer** | The outermost layer of the endosperm in a maize grain, highly **rich in protein**. |
| **Scutellum** | The single, large **cotyledon** of a monocot embryo (like maize). |

| Biological Term 🔑 | Exam-Ready Definition 📝 |
|---|---|
| **Coleorhiza** | A protective sheath enclosing the **radicle** in a monocot seed. |
| **Coleoptile** | A protective sheath enclosing the **plumule** in a monocot seed. |
| **Germination** | The process of formation of a **seedling** developed from the embryo under proper conditions. |
| **Hypogeal Germination** | Germination where the epicotyl elongates faster, and cotyledons **remain underground**. |
| **Epigeal Germination** | Germination where the hypocotyl elongates faster, pushing cotyledons **above the ground**. |
| **Vivipary (Viviparous)** | A special mode where the seed germinates **inside the fruit** while still attached to the parent plant. |
| **Seedling** | A stage in the growth of a plant from a seed before it becomes **wholly independent** of its stored food. |

## 🌟 2. Master Example List 🌿

- **Seed & Fruit Structures:**
  - *Fruit (Separate Seed):* Mango, pea pod.
  - *Seed:* Bean seed, peas.
  - *Grain (Fused Wall/Coat):* Maize, wheat, rice.
- **Seed Size Extremes:**
  - *Barely visible:* Poppy seeds, strawberry seeds.

- *Quite large:* Mango, plum.
- *Largest seeds:* Coconut, double coconut.

- **Seed Classification by Cotyledons & Endosperm:**
  - *Monocot Albuminous:* Cereals, millets, palm.
  - *Dicot Albuminous:* Poppy, custard apple.
  - *Monocot Exalbuminous:* Vallisneria, orchids, amorphophallus.
  - *Dicot Exalbuminous:* Gram, pea, mango, mustard.
  - *Dicot Seeds* – Castor, Bean
  - *Monocot Seeds* – Grasses,Maize
- **Types of Germination:**
  - *Hypogeal Germination:* Pea, gram, maize.
  - *Epigeal Germination:* Castor, bean.
  - *Viviparous Germination:* Mangrove plants (Rhizophora, Sonneratia).
- **Roots:**
  - *Tap Root:* Found typically in dicotyledonous plants.
  - *Fibrous Root:* Found typically in monocotyledonous plants.
  - *Adventitious Roots:* Develop from the base of the stem in maize after the radicle dies off.

### ⚙️ 3. Master Function & Location Matrix 📍🛠️

| **Biological Part** 💉 | **Exact Anatomical Location** 📍 | **Primary Function** ⚙️ |
|---|---|---|
| **Fruit Wall** | Outermost covering of the fruit. | Encloses seeds, protects them, and helps in seed dispersal. |

| Biological Part 🧬 | Exact Anatomical Location 📍 | Primary Function ⚙️ |
|---|---|---|
| **Seed Coat** | Outermost covering of the seed (Testa + Tegmen). | Protects the embryo from mechanical damage, bacteria, fungi, and insects. |
| **Endosperm** | Inside albuminous seeds (e.g., majority part of maize grain). | Stores food material for the nourishment of the embryo during germination. |
| **Cotyledons** | Attached to the embryo axis below the seed coat. | Store food (in exalbuminous seeds), supply food to the seedling, and may form first green leaves. |
| **Micropyle** | Tiny pore close to the hilum on the seed coat. | Absorbs water for germination; provides for diffusion of respiratory gases. |
| **Hilum** | Whitish oval scar on the concave side of the seed. | Point of attachment to the placenta/ovary wall. |
| **Epithelial Layer** | Between the endosperm and embryonic part in maize. | Separates the endosperm from the embryo. |
| **Aleurone Layer** | Outermost layer of the endosperm in maize. | Highly rich in protein. |

| Biological Part 🧬 | Exact Anatomical Location 📍 | Primary Function ⚙️ |
|---|---|---|
| **Scutellum** | One side of the maize grain (the single cotyledon). | Absorbs food from the endosperm till it is exhausted during germination. |
| **Coleorhiza** | Lower pointed end of the maize embryo axis. | Protective sheath enclosing the radicle. |
| **Coleoptile** | Upper broader side of the maize embryo axis. | Protective sheath enclosing the plumule; pierces soil during germination. |
| **Alkaline Pyrogallic Acid** | Inside the test tube (Experiment setup). | Absorbs oxygen to prove its necessity for germination. |
| **Plain Water (Test Tube)** | Inside the control test tube (Experiment setup). | Maintains a humid environment. |

❓ **Progress Check (True/False)**

1. **Mention whether the following statements are True (T) or False (F):**

- **(i)** Plumule is the future root and radicle is the future shoot of the plant. ➡️ **False** (F)
- **(ii)** Micropyle serves for the emergence of the radicle. ➡️ **False** (F)
- **(iii)** Cotyledons in castor are a big store of food for the embryo. ➡️ **False** (F)
- **(iv)** Maize grain has a large endosperm. ➡️ **True** (T)
- **(v)** Tegmen is the outermost layer of the seed. ➡️ **False** (F)

## ❓ Progress Check (Fill in the blanks)

**1. Fill in the blanks by choosing the correct alternative:**

- **(i)** In a dry seed, the embryo is in **quiescent** state.
- **(ii)** Water is absorbed by the seed mainly through **micropyle**.
- **(iii)** Very low temperature **inhibits** the growth of embryo.
- **(iv)** Germination in pea is **hypogeal**.
- **(v)** Coleoptile is a part found in the germinating **maize**.
- **(vi)** Alkaline pyrogallic acid is used for absorbing **oxygen** during experiments on germination.

## 📝 Review Questions: A. MULTIPLE CHOICE TYPE

**1. Choose the correct answer from the options given below:**

- **(i)** Which one of the following plant parts is correctly matched with one of its stated characteristic? ➡️ **(c) Maize grain ..... coleoptile**
- **(ii)** Seeds sown very deep in the soil fail to germinate because they: ➡️ **(d) do not get enough oxygen.**
- **(iii)** Fruit is a ripened: ➡️ **(a) Ovary**
- **(iv)** Which of the following is the largest seed? ➡️ **(c) Double coconut**
- **(v)** The distinct whitish oval scar on the concave side of the seed is termed as: ➡️ **(c) Hilum**
- **(vi)** The single cotyledon of a maize grain is called: ➡️ **(b) Scutellum**
- **(vii)** The outermost hard brownish layer of the seed coat is: ➡️ **(a) Testa**
- **(viii)** The part of an embryo which develops into root is: ➡️ **(d) Radicle**
- **(ix)** Which of the following statement holds true for a maize grain? ➡️ **(d) All of the above.**
- **(x)** Seema observed different kinds of seeds... Which of her above observations match with bean seeds? ➡️ **(d) P and R**

- **(xi)** Seeds germinate in two ways. The type in which the cotyledons remain underground is called (P) ..... germination. During this type of germination, the (Q) ..... elongates faster. ➡️ **(d) P – Hypogeal, Q – Epicotyl**

## ✍️ B. VERY SHORT ANSWER TYPE

**1. Correct the following false statements by changing the first/last word only.**

- **(a)** Plumule develops into the root system. ➡️ **Radicle** develops into the root system.
- **(b)** Testa is the thin inner layer of the seed coat. ➡️ **Tegmen** is the thin inner layer of the seed coat.
- **(c)** Endothelium separates the endosperm and embryonic region in the maize grain. ➡️ **Epithelium** separates the endosperm and embryonic region in the maize grain.
- **(d)** Hypocotyl elongates faster in hypogeal germination. ➡️ **Epicotyl** elongates faster in hypogeal germination.
- **(e)** Germination is the period of rest in a seed. ➡️ **Dormancy** is the period of rest in a seed.

**2. Name the following:**

- **(a)** A monocotyledonous endospermic seed. ➡️ Maize (or rice/wheat) grain.
- **(b)** A chemical used in experiments, which absorbs oxygen. ➡️ Alkaline pyrogallic acid.
- **(c)** Part of the plumule above the embryonic axis of the seed. ➡️ Epicotyl.
- **(d)** A plant which shows viviparous germination. ➡️ Rhizophora (or Sonneratia).
- **(e)** The layer of endosperm of maize, rich in protein. ➡️ Aleurone layer.
- **(f)** A seed with folded plumule leaves. ➡️ Bean seed.

**3. Give two examples of each of the following:**

- **(a)** Monocot albuminous seed. ➡️ Cereals, millets (e.g., maize, palm).
- **(b)** Dicot albuminous seed. ➡️ Poppy, custard apple.
- **(c)** Monocot non-endospermic seed. ➡️ Vallisneria, orchids.
- **(d)** Dicot non-endospermic seed. ➡️ Gram, pea.
- **(e)** Viviparous plant. ➡️ Rhizophora, Sonneratia.

**4. Match the structures marked (a) to (e) with their correct functions/characteristics (L.S. of maize grain):**

- **(a)** ➡️ 2. Single cotyledon.
- **(b)** ➡️ 1. Develops into the root system.
- **(c)** ➡️ 3. Later forms the shoot.
- **(d)** ➡️ 5. Rich in starch.
- **(e)** ➡️ 4. Rich in protein.
- **(f)** ➡️ 6. Separates the endosperm and the embryonic part.

**5. Fill in the blanks:**

- **(a)** In bean seeds, **hypocotyl** grows faster and the seeds are brought **above** ground.
- **(b) Coleorhiza** is a protective layer of radicle and **coleoptile** protects the rolled plumule.
- **(c)** A seed is protected by **testa** and **tegmen**.
- **(d)** Seeds absorb water through **micropyle** which also helps in diffusion of respiratory gases.
- **(e)** Rice, wheat, and maize are rich in **starch** food.

**6. Arrange the following set of terms in order, so as to be in logical sequence:**

- **(a)** Embryo, 1st male gamete, zygote, egg cell, micropyle. ➡️ Micropyle, 1st male gamete, egg cell, zygote, embryo.

- **(b)** Zygote, embryo, seed, allogamy, fusion of gametes. ➡️ Allogamy, fusion of gametes, zygote, embryo, seed.
- **(c)** Tegmen, Embryonic axis, Cotyledons, Testa (arrange with respect to their position in a seed from outer to inner). ➡️ Testa, Tegmen, Cotyledons, Embryonic axis.

**7. Match the terms given in column A with those of column B:**

- **(a)** Coleoptile ➡️ (iv) Plumule
- **(b)** Coleorhiza ➡️ (iii) Radicle
- **(c)** Radicle ➡️ (i) Root system
- **(d)** Plumule ➡️ (v) Shoot system
- **(e)** Scutellum ➡️ (ii) Cotyledon

## 📋 C. SHORT ANSWER TYPE

**8. Name the following:**

- **(a)** Two types of seeds on the basis of the number of cotyledons. ➡️ Monocotyledonous and Dicotyledonous seeds.
- **(b)** Two types of seeds on the basis of endosperm. ➡️ Albuminous (endospermic) and Exalbuminous (non-endospermic) seeds.
- **(c)** Three necessary conditions for the germination of seeds. ➡️ Water, Suitable (optimum) temperature, and Oxygen.
- **(d)** Two main types of germination. ➡️ Epigeal and Hypogeal germination.
- **(e)** Two layers of the seed coat. ➡️ Testa and Tegmen.

## 📘 C. SHORT ANSWER TYPE

**1. Define the following terms:**

- **(a) Seed:** A seed is the ripened ovule containing an embryo which develops into a new plant.
- **(b) Fruits:** A fruit is the enlarged, ripened ovary whose wall encloses and protects the seeds.

- **(c) Grain:** A grain is actually a single-seeded fruit in which the fruit wall and the seed coat are completely fused together to form a protective layer.
- **(d) Dormancy:** A period of rest during which the embryo inside the seed remains in an inactive state until exposed to favourable conditions.
- **(e) Germination:** The process of formation of a seedling developed from the embryo when placed under proper conditions.

**2. What is the difference between an embryo and a seed?**

- A **seed** is the entire ripened ovule, consisting of the seed coat, stored food (cotyledons/endosperm), and the embryo.
- An **embryo** is just the tiny living plant *inside* the seed (consisting of the radicle, plumule, and embryonic axis) that actually grows into the new plant.

**3. Germinated grams are considered highly nutritive. What is the reason for this belief?**

- During germination, enzymes become active and convert the complex, insoluble stored food (like starch and proteins) into simple, soluble, and easily digestible forms (like sugars and amino acids). Additionally, germination synthesizes new vitamins (especially Vitamin C), making them highly nutritive.

**4. Why do we not use the terms maize fruit and maize seed? What do we say instead?**

- We do not use those terms because in maize, the fruit wall and the seed coat are completely fused together, making it impossible to separate the seed from the fruit. Instead, we use the term **grain**.

## 📙 D. LONG ANSWER TYPE

**1. Distinguish between the following pairs:**

- **(a) Monocotyledonous and Dicotyledonous seeds:** Monocot seeds contain only one cotyledon (e.g., maize), while dicot seeds contain two cotyledons (e.g., bean).

- **(b) Epicotyl and Hypocotyl:** Epicotyl is the region of the embryo axis *above* the point of attachment of cotyledons (between cotyledons and plumule). Hypocotyl is the region *below* the cotyledons (between cotyledons and radicle).
- **(c) Epigeal and Hypogeal germination:** In epigeal germination, the hypocotyl elongates faster, pushing the cotyledons *above* the ground. In hypogeal germination, the epicotyl elongates faster, and the cotyledons remain *underground*.
- **(d) Radicle and Plumule:** The radicle is the part of the embryo that develops into the future *root* system. The plumule develops into the future *shoot* system.
- **(e) Albuminous and Exalbuminous seeds:** Albuminous seeds possess a persisting endosperm to store food (cotyledons are thin). Exalbuminous seeds consume the endosperm fully, and the food is stored in thick, fleshy cotyledons.

**2. What are the functions of the following in a seed?**

- **(a) Seed coat:** Protects the delicate inner parts of the embryo from mechanical injury, bacteria, fungi, and insects.
- **(b) Micropyle:** Absorbs water for germination and provides a passageway for the diffusion of respiratory gases for the growing embryo.
- **(c) Cotyledons:** Store food for the embryo (in exalbuminous seeds) and provide nourishment to the seedling until it can photosynthesize.
- **(d) Radicle:** Elongates and grows downwards into the soil to form the root system, absorbing water and minerals.
- **(e) Plumule:** Grows upwards above the soil to form the shoot system (stem and leaves).

**3. Suggest an experiment to prove that a suitable temperature is necessary for germination.**

- Take two beakers (A and B) and place some green gram seeds on wet cotton wool in both. Keep Beaker A in an ordinary room (optimum temperature) and Beaker B inside a refrigerator (very low temperature).

After a few days, seeds in Beaker A will germinate, while seeds in Beaker B will not show signs of germination, proving that a suitable, moderately warm temperature is necessary.

**4. Sometimes the potatoes kept in a basket during the late rainy season start giving out small shoots. Would you call it germination? Give reason in support of your answer.**

- **No**, this is not called germination.
- **Reason:** Germination strictly refers to the development of a seedling from an *embryo inside a seed* (a ripened ovule). A potato is a modified underground stem (tuber), not a seed. The sprouting of shoots from its "eyes" (buds) is a form of vegetative propagation, not seed germination.

**5. Give two differences in each of the following pairs:**

- **(a) Coleorhiza and coleoptile:** Coleorhiza is a protective sheath enclosing the *radicle* at the lower end. Coleoptile is a protective sheath enclosing the *plumule* at the upper end.
- **(b) Bean seed and maize grain:** Bean seed has two cotyledons and no endosperm (exalbuminous). Maize grain has one cotyledon and a large endosperm (albuminous).
- **(c) Germination and vivipary:** Germination is the general process of a seed developing into a seedling when sown in soil. Vivipary is a special type of germination where the seed germinates *inside the fruit* while it is still attached to the parent plant.

**6. Justify the statement that the maize grain is a 'one seeded fruit'.**

- In maize, the outer covering is not just a seed coat. It is formed by the complete fusion of the fruit wall (pericarp) and the seed coat (testa/tegmen). Because it develops from a ripened ovary containing a single ovule, and the walls are inseparable, it is scientifically a one-seeded fruit.

**7. What is the role played by the hypocotyl in epigeal germination?**

- In epigeal germination, the hypocotyl elongates rapidly and forms an arch/loop above the soil. As it straightens, it physically pulls and pushes the cotyledons and plumule above the ground.

*(Questions 8 & 9 from Image 2 ask for drawings. As an AI, I cannot draw on paper, but you should practice sketching the 'Viviparous twig' and the 'Three-bean experiment' directly from your textbook!)*

## 🟢 E. STRUCTURED/APPLICATION/SKILL TYPE

**1. The figure shown below represents the internal structure of a certain seed. Study the figure and answer the following questions.**

- **(a) Identify the seed and mention whether it is mono or dicotyledonous.**
    - It is a **Bean seed**. It is **dicotyledonous** (has two cotyledons).
- **(b) Label the guidelines 1 to 4.**
    - 1: Plumule
    - 2: Radicle
    - 3: Epicotyl
    - 4: Cotyledon
- **(c) Write an important role of part 2.**
    - Part 2 (Radicle) grows downwards into the soil to form the root system.
- **(d) Mention the food component found in part 4.**
    - Part 4 (Cotyledon) is primarily rich in proteins and carbohydrates (starch) to nourish the embryo.
- **(e) What kind of germination takes place in the above mentioned seed?**
    - **Epigeal germination** (the cotyledons are pushed above the ground).

**2. Kiara visited a forest with marshy land. She observed that in some of the plants, the seeds germinate inside the fruit while still being attached to the parent plant...**

- **(a) What kind of germination is shown above in the figure?**
    - **Viviparous germination** (Vivipary).
- **(b) Give two examples of the plants which germinate by this kind of germination.**
    - Rhizophora and Sonneratia (Mangroves).
- **(c) Mention the part shown in the figure as 'A'.**
    - **Hypocotyl** (In viviparous plants, the hypocotyl elongates heavily downwards to act as a heavy dart to pierce the mud).

**3. With regard to germination in bean seed, answer the following questions:**

- **(a) State the function of the 'Micropyle'.**
    - It absorbs water to swell the seed and provides a passage for the diffusion of respiratory gases.
- **(b) Name the part of the seed that grows into the seedling.**
    - The **Embryo**.
- **(c) Name the part of the seed that provides nutrition for the growing seedling.**
    - The **Cotyledons**.
- **(d) Draw a neat labelled diagram of a mature bean seed.**
    - *(Practice sketching the external view showing the Testa, Hilum, and Micropyle!)* 🖍️

## 🧩 F. ASSERTION - REASON TYPE QUESTIONS

*(Codes: (a) Both True, R explains A. (b) Both True, R doesn't explain A. (c) A True, R False. (d) A False, R True)*

- **(i) Assertion (A) : Fruit is the enlarged ripened ovule which protects the seeds. Reason (R) : Fruits develop from the ovary...**
    - **Answer: (d) A is false but R is true.** *(Fruit is a ripened ovary, not ovule).*

- **(ii) Assertion (A) : Dicotyledonous seeds are always exalbuminous... Reason (R) : All dicotyledonous seeds contain two cotyledons which store food...**
    - **Answer: (d) A is false but R is true.** *(Note: A is definitely false because some dicots like poppy are albuminous. The ICSE board key typically marks R as true in a general structural sense).*
- **(iii) Assertion (A) : Micropyle is the opening through which the pollen tube enters the ovule. Reason (R) : Micropyle helps in the diffusion of respiratory gases...**
    - **Answer: (b) Both A and R are true but R is not the correct explanation of A.** *(Both facts are biologically correct, but one is a pre-fertilization function and the other is a germination function).*
- **(iv) Assertion (A) : The single cotyledon of maize grain is termed as scutellum which does not store any food... Reason (R) : Food material... is stored in the endosperm...**
    - **Answer: (a) Both A and R are true and R is the correct explanation of A.**
- **(v) Assertion (A) : Epicotyl of the embryonic axis elongates faster in hypogeal germination. Reason (R) : Epicotyl of the embryonic axis elongates faster in epigeal germination and the cotyledons remain underground.**
    - **Answer: (c) A is true but R is false.** *(Reason is totally mixed up; hypocotyl elongates in epigeal).*

🔬 **G. CASE STUDY BASED QUESTION**

**1. Vinita has gone to her grandparents' home situated in a 'coastal village'...**

- **(a) Mention the special mode of seed germination seen by Vinita in some plants of the coastal area.**
    - **Viviparous germination** (Vivipary).
- **(b) What specific term is used to describe such plants in which the seed germinates inside the fruit itself?**

- **Mangrove plants** (or Viviparous plants).

- **(c) Give two examples of the plants described above in (b).**
  - Rhizophora and Sonneratia.
- **(d) Which part of the embryo appears much elongated during the above mentioned germination in (a)?**
  - The **Hypocotyl**.
- **(e) How are these seeds adversely affected if they fall directly on to the marshy land?**
  - If they fell as normal seeds, they would suffocate due to the lack of oxygen in the waterlogged, salty mud. They would also easily be washed away by the harsh ocean tides before they could take root. Vivipary ensures they fall as heavy, pre-sprouted darts that instantly anchor into the mud.